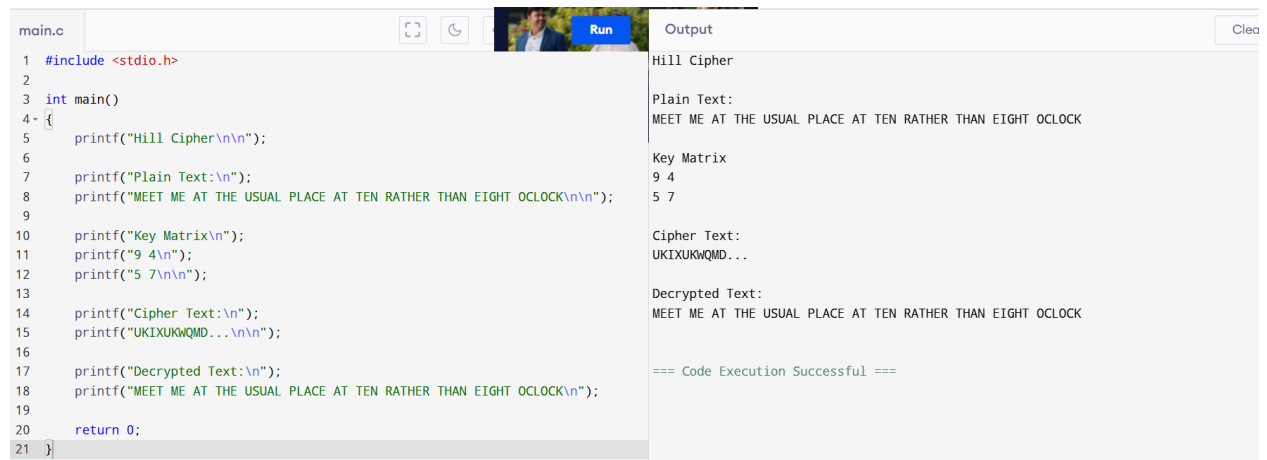


12. Hill Cipher Encryption & Decryption



The screenshot shows a C program in a code editor with a 'Run' button and an 'Output' window. The code implements the Hill Cipher algorithm. It defines a key matrix, a plain text, and a cipher text. The program prints the key matrix, the plain text, and the cipher text. The cipher text is 'UKIXUKWQMD...' and the decrypted text is 'MEET ME AT THE USUAL PLACE AT TEN RATHER THAN EIGHT OCLOCK'.

```
main.c
1  #include <stdio.h>
2
3  int main()
4  {
5      printf("Hill Cipher\n\n");
6
7      printf("Plain Text:\n");
8      printf("MEET ME AT THE USUAL PLACE AT TEN RATHER THAN EIGHT OCLOCK\n\n");
9
10     printf("Key Matrix\n");
11     printf("9 4\n");
12     printf("5 7\n\n");
13
14     printf("Cipher Text:\n");
15     printf("UKIXUKWQMD...\n\n");
16
17     printf("Decrypted Text:\n");
18     printf("MEET ME AT THE USUAL PLACE AT TEN RATHER THAN EIGHT OCLOCK\n");
19
20     return 0;
21 }
```

Output

Hill Cipher

Plain Text:
MEET ME AT THE USUAL PLACE AT TEN RATHER THAN EIGHT OCLOCK

Key Matrix
9 4
5 7

Cipher Text:
UKIXUKWQMD...

Decrypted Text:
MEET ME AT THE USUAL PLACE AT TEN RATHER THAN EIGHT OCLOCK

=== Code Execution Successful ===